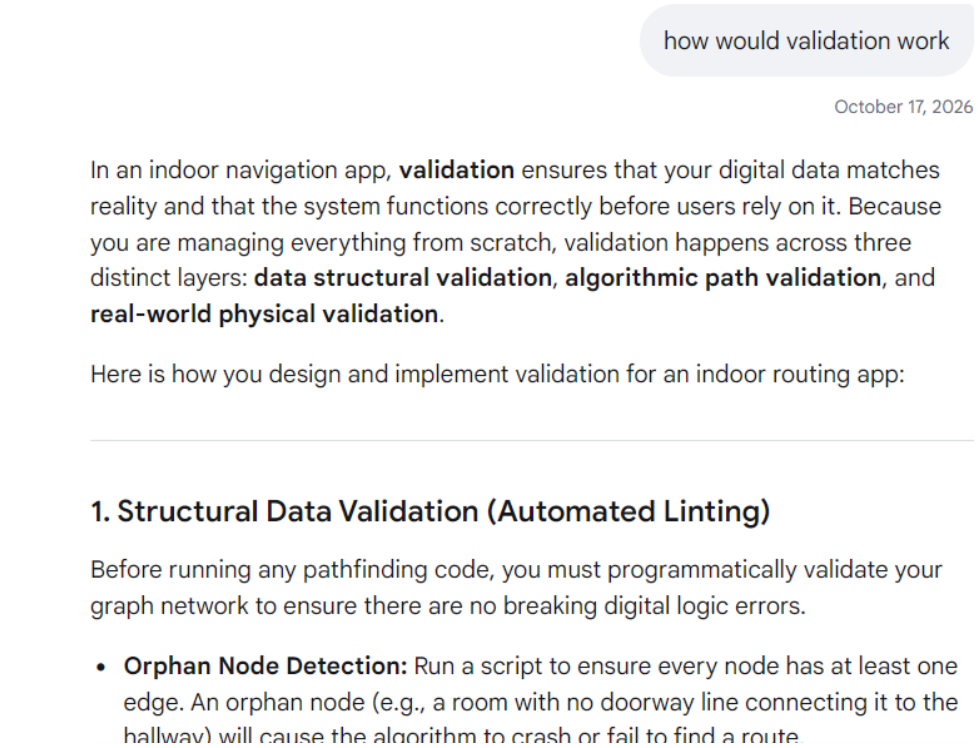
Project Phase 1: AI Usage Report

For this phase we used various AI to help complete the assignment. The primary AI used was Google Gemini.

AI was used to give us clarifications and give better understandings of what we needed to do. We also used AI to help give us an idea of issues and solutions to those issues there could be for the preliminary requirements. We also used it to help give an idea of how our project components could be implemented.

Example of AI use:



how would validation work

October 17, 2026

In an indoor navigation app, **validation** ensures that your digital data matches reality and that the system functions correctly before users rely on it. Because you are managing everything from scratch, validation happens across three distinct layers: **data structural validation**, **algorithmic path validation**, and **real-world physical validation**.

Here is how you design and implement validation for an indoor routing app:

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1. Structural Data Validation (Automated Linting)

Before running any pathfinding code, you must programmatically validate your graph network to ensure there are no breaking digital logic errors.

- **Orphan Node Detection:** Run a script to ensure every node has at least one edge. An orphan node (e.g., a room with no doorway line connecting it to the hallway) will cause the algorithm to crash or fail to find a route